

Sir Syed University of Engineering and Technology
Department of Computer Science and Information Technology
CS-326 Information Security
ASSIGNMENT 01

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Section: A

Max Marks: 03
Obtained Marks:
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Q#1: Consider that an individual threat agent, like a hacker, can be a factor in more than one threat category. If a hacker breaks into a network, copies a few files, defaces a Web page, and steals credit card numbers, how many different threat categories does the attack fall into?

This attack falls into four different threat categories, based on the actions performed by the hacker. Let's break it down:

1. **Espionage or Trespass:**
 - The hacker **breaks into a network** and **copies files**.
 - This is unauthorized access and surveillance, which fits the espionage/trespass category.
2. **Information Extortion (or Theft):**
 - The **theft of credit card numbers** is a classic case of stealing sensitive information for malicious purposes, often followed by threats or blackmail.
3. **Sabotage or Vandalism:**
 - **Defacing a web page** is an act of vandalism that harms the integrity and image of the organization.
4. **Compromises to Intellectual Property** (depending on the files copied):
 - If the copied files include proprietary information, then this could also fall into intellectual property threats.

However, **at minimum**, the attack clearly fits into **three primary categories**:

- **Espionage/Trespass**
- **Sabotage/Vandalism**
- **Information Theft/Extortion**

So, **3 to 4 categories** depending on the content of the stolen files.

Q#2: Using the Web, research Mafiaboy's exploits. When and how did he compromise sites? How was he caught?

In early 2000, a 15-year-old Canadian hacker known by the alias **Mafiaboy** (real name Michael Calce) orchestrated a series of high-profile cyberattacks that disrupted major websites and exposed significant vulnerabilities in internet security.

How and When He Compromised Sites

Between February 7 and February 14, 2000, Mafiaboy launched **Distributed Denial-of-Service (DDoS)** attacks against several prominent websites, including:

- **Yahoo!**
- **CNN.com**
- **Amazon.com**
- **eBay**
- **Dell**
- **E*TRADE**
- **FIFA.com**

These attacks involved overwhelming the target websites with massive amounts of traffic, rendering them inaccessible to legitimate users. Mafiaboy achieved this by compromising numerous computers to create a botnet, which he then used to flood the targeted sites with data, effectively crashing their servers.

How He Was Caught

Mafiaboy's downfall began when he **boasted about his exploits in online chat rooms**, inadvertently drawing attention to himself. Investigators traced the origin of the attacks to computers at the **University of California at Santa Barbara**, which had been used as part of the botnet. Further investigation led authorities to a computer at Mafiaboy's high school in Montreal. He was arrested in April 2000 and charged with multiple counts of "mischief to data."

Legal Outcome

In January 2001, Mafiaboy pleaded guilty to **55 charges** related to the cyberattacks. He was sentenced to **eight months in a youth detention center**, followed by one year of probation and a fine of \$160. The attacks were estimated to have caused damages ranging from **\$7.5 million to \$1.7 billion**, highlighting the significant impact of his actions.

Mafiaboy's case served as a wake-up call for the importance of cybersecurity and the potential threats posed by individuals exploiting vulnerabilities in internet infrastructure.

Q#3: Using the categories of threats mentioned in chapter 02 and the various attacks described, review several cAurrent media sources and identify examples of each threat.

Cybersecurity threats in 2025 have evolved significantly, driven by advancements in technology, geopolitical tensions, and increasingly sophisticated cybercriminal tactics. Here's an overview of the most pressing threats and trends shaping the digital security landscape this year:

1. Sophisticated Ransomware Attacks

Ransomware remains a dominant threat, with attackers employing advanced techniques such as double extortion—encrypting data and threatening to release sensitive information unless a ransom is paid. Critical infrastructure, healthcare systems, and financial institutions are primary targets, highlighting the need for robust cybersecurity measures.

2. AI-Enhanced Cybercrime

Artificial intelligence is increasingly utilized by cybercriminals to automate and enhance attacks. AI-driven tactics include sophisticated phishing campaigns, deepfake impersonations, and rapid exploitation of vulnerabilities, making detection and defense more challenging.

3. State-Sponsored Cyber Espionage

Nation-state actors continue to pose significant threats:

- **CyberAv3ngers:** An Iranian group targeting industrial control systems in sectors like water and energy, using custom malware to disrupt operations.
- **Brass Typhoon (APT41):** A Chinese-linked group conducting espionage across various industries, employing sophisticated malware and supply chain attacks.
- **Salt Typhoon:** Infiltrated U.S. telecom networks, accessing sensitive metadata and communications of high-profile individuals, indicating a focus on surveillance and intelligence gathering.

4. Quantum Computing Threats

The advancement of quantum computing poses a future risk to current encryption standards. Efforts are underway to develop quantum-resistant cryptographic solutions to safeguard data against potential quantum-enabled breaches.

5. Supply Chain Vulnerabilities

Cyberattacks targeting supply chains have increased, exploiting interconnected systems to gain access to multiple organizations. These attacks can have widespread impacts, emphasizing the importance of securing third-party relationships.

Strategic Recommendations

To mitigate these evolving threats, organizations should consider the following strategies:

- **Implement AI-Driven Security Solutions:** Utilize artificial intelligence for real-time threat detection and response.
- **Adopt Zero Trust Architecture:** Ensure strict verification for all users and devices attempting to access network resources.
- **Enhance Supply Chain Security:** Conduct thorough assessments of third-party vendors and enforce stringent security standards.
- **Prepare for Quantum Threats:** Begin transitioning to quantum-resistant encryption methods to future-proof sensitive data.
- **Invest in Employee Training:** Educate staff on recognizing and responding to phishing attempts and other social engineering tactics.

Staying informed and proactive is crucial in navigating the complex cybersecurity landscape of 2025.

Continuous adaptation and investment in advanced security measures will be key to protecting assets and maintaining trust in an increasingly digital world.